Master Theory A-Z

Linear Regression, Regularization, GLMs, Logistic Regression and diagnostics explained simply

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How to read this theory guide

This guide explains the core concepts in simple language first, then connects them to the notebooks and business use cases. It is not only a definition list. It explains why each concept matters and how it changes analytical decisions.

> **Simple mental model**
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> Regression is like drawing a useful rule through messy points. Classification is like turning customer signals into a probability of belonging to a class. Regularization is like telling the model: do not over-trust every feature. GLM is like changing the model so it respects the shape of the target, such as counts.

Supervised learning, regression and classification

Supervised learning uses examples where the input and the correct output are already known. In regression the output is a number, such as Sales, house price or rental count. In classification the output is a class, such as responder vs non-responder. Day 12 uses both ideas: Linear Regression predicts numbers, while Logistic Regression predicts probabilities for classes.

Linear Regression

Linear Regression models a straight-line relationship between inputs and a numeric target. With one feature, the model is y = intercept + coefficient * x. With many features, it adds one coefficient for each feature. The coefficient tells the expected direction and size of impact when that feature increases and the other features stay fixed.

OLS and least squares

OLS means Ordinary Least Squares. It is not the same thing as the idea of linear regression; it is one common method used to estimate the coefficients. It chooses coefficients that minimize the sum of squared residuals. A residual is actual value minus predicted value. Squaring makes large mistakes more expensive and avoids positive and negative errors cancelling out.

Closed-form OLS vs gradient descent

Closed-form OLS uses matrix algebra to directly solve for the best coefficients when the problem is small and well-behaved. Gradient descent starts with guesses and improves them step by step by moving in the direction that reduces error. Closed-form is like solving the equation directly; gradient descent is like walking downhill until the loss cannot get much lower.

Correlation vs regression

Correlation measures how strongly two numeric variables move together. Regression builds a predictive equation with a target and features. Correlation is symmetric; regression is directional. TV and Sales can be correlated, but regression asks: how much Sales changes when TV budget changes.

Interpolation vs extrapolation

Interpolation means predicting inside the range of data the model has already seen. Extrapolation means predicting outside that range. Extrapolation is risky because a straight-line pattern observed in the data may not continue forever.

Regression assumptions

Linear regression is easiest to trust when relationships are roughly linear, residuals are independent, errors have similar spread, and extreme observations do not dominate the line. These are assumptions to check, not blind rules to memorize.

Homoscedasticity and heteroscedasticity

Homoscedasticity means the residual spread is roughly constant across prediction levels. Heteroscedasticity means the errors get wider or narrower as predictions change. This matters because unreliable error spread can make confidence intervals and p-values less trustworthy.

Outliers, leverage and Cook distance

An outlier has an unusual target value. A leverage point has unusual feature values. Cook distance combines both ideas and asks whether a row has too much influence on the fitted model. A high Cook distance row is not automatically wrong, but it deserves investigation.

Ridge, Lasso and ElasticNet

Regularization adds a penalty to reduce unstable coefficients. Ridge shrinks coefficients but usually keeps all features. Lasso can set weak features to exactly zero. ElasticNet mixes both ideas and is useful when features are correlated but feature selection is also desired.

Cross-validation and LassoCV

A single train/test split can be lucky or unlucky. Cross-validation repeats the validation idea across multiple folds. LassoCV uses this to choose alpha, the regularization strength, more fairly than guessing one value.

GLM, Poisson and Negative Binomial

A Generalized Linear Model changes the distribution and link function so the model fits the target type better. Poisson is designed for counts when the mean and variance are similar. Negative Binomial is often better when the count target is overdispersed, meaning the variance is much larger than the mean.

Logistic Regression

Logistic Regression is a classification model, despite its name. It creates a linear score, then passes it through the sigmoid function to produce a probability between 0 and 1. A threshold converts probability into a class decision. This makes it useful for churn, response, fraud and lead scoring.

sklearn vs statsmodels

scikit-learn is prediction-workflow focused: pipelines, train/test split, cross-validation and scoring. statsmodels is inference focused: p-values, confidence intervals, robust standard errors, AIC/BIC and model summaries. A professional analyst often uses both: sklearn for production-style prediction, statsmodels for explanation and statistical interpretation.

Concept map

Concept	Plain meaning	Notebook evidence
OLS	Find coefficients by minimizing squared residuals.	Notebooks 1 and 5
Gradient descent	Learn by reducing loss step by step.	Notebook 5
Ridge	Shrink all coefficients.	Notebook 2
Lasso	Shrink and remove weak features.	Notebook 2
ElasticNet	Blend Ridge and Lasso.	Notebook 2
Poisson	Regression for count outcomes.	Notebook 3
Negative Binomial	Count model for overdispersion.	Notebook 3
Logistic Regression	Probability model for classification.	Notebook 4

References and source foundation

The documents use the completed Day 12 notebooks as the main project evidence. The theory explanations are also aligned with the uploaded course materials and official library documentation.

Source	How it was used
Uploaded course guides: leit03.pdf and Haupt_leitfaden03.pdf	OLS, least squares, residuals, regression line, correlation, interpolation/extrapolation.
Uploaded ML slides: Methode der Kleinsten Quadrate and Typen des Machinellen Lernens	Supervised learning, regression vs classification, gradient descent, logistic regression context.
scikit-learn Linear Models User Guide - https://scikit-learn.org/stable/modules/linear_model.html	Linear Regression, Ridge, Lasso, ElasticNet, Logistic Regression, PoissonRegressor.
scikit-learn Model Evaluation User Guide - https://scikit-learn.org/stable/modules/model_evaluation.html	MSE, MAE, R2, ROC-AUC, PR-AUC, precision, recall, F1, calibration.
statsmodels GLM documentation - https://www.statsmodels.org/stable/glm.html	Generalized Linear Models, Poisson family and model interpretation.
statsmodels NegativeBinomial documentation - https://www.statsmodels.org/stable/generated/statsmodels.discrete.discrete_model.NegativeBinomial.html	Negative Binomial implementation details and count-model context.
UCLA IDRE Negative Binomial Regression notes - https://stats.oarc.ucla.edu/r/dae/negative-binomial-regression/	Practical explanation of Negative Binomial as a model for overdispersed count outcomes.